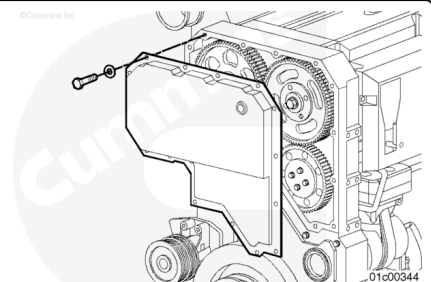


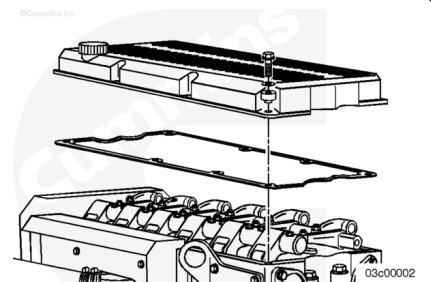
Trainings ▾

Preparatory Steps

Remove the upper gear cover. Refer to Procedure 001-079 (</qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html>) in Section 1.

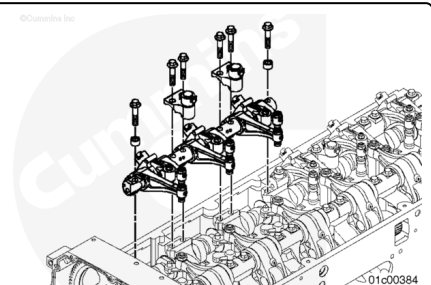


Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>).

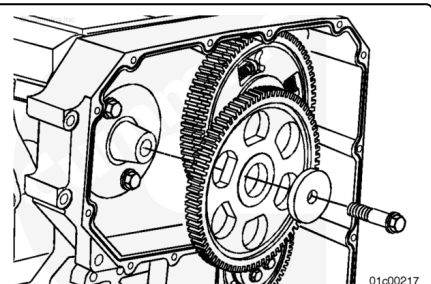


Disconnect the engine brake wiring harness. Refer to Procedure 020-015 in Section 20 (</qs3/pubsys2/xml/en/procedures/10/10-020-015.html>).

Remove the rocker lever shafts. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html>) in Section 3.



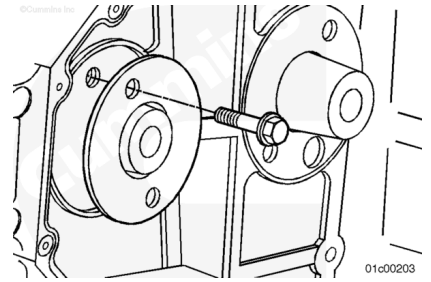
Remove the valve camshaft gear. Refer to Procedure 001-088 (</qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html>) in Section 1.



Remove

Remove the retainer capscrews, retainer, seal, and o-ring.

Discard the seal and o-ring.

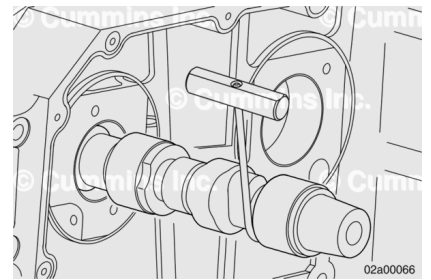


⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ CAUTION ⚠

Use caution when removing the camshaft to avoid damaging the camshaft bushings.



Use the Valve Camshaft Removal and Installation Kit, Part Number 5299837, to remove the camshaft from the cylinder head.

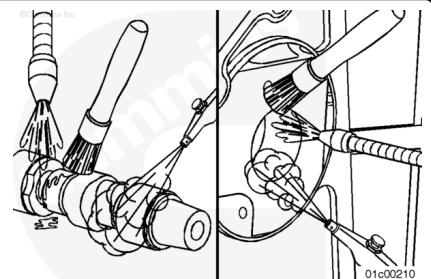
Use the Camshaft Removal and Installation Kit, Part Number 5395348, to pull and guide the camshaft out of the cylinder head until three journals remain in the cylinder head. Carefully remove the remainder of the camshaft by hand.

Clean and Inspect for Reuse

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ WARNING ⚠



When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

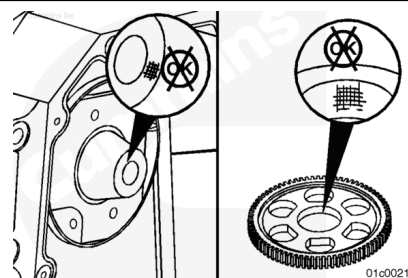
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the valve camshaft and the bore of the valve camshaft gear with ENVIROSOL™ 655 solvent, or equivalent, and a lint-free cloth.

Clean all debris from the cylinder head and the camshaft bushings.

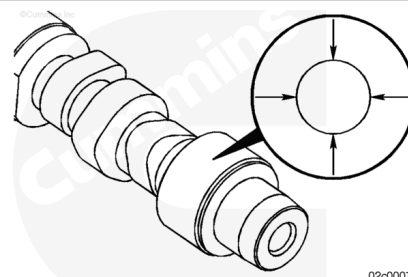
Check the valve camshaft and valve camshaft gear for fretting damage.

The valve camshaft and valve camshaft gear **must** be replaced if damaged.



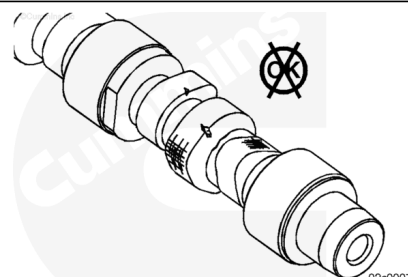
Measure the outside diameter of the seven valve camshaft bearing journals.

Valve Camshaft Bearing Journal Outside Diameter		
mm		in
64.987	MIN	2.5585
65.013	MAX	2.5596



If the camshaft journals are **not** within specifications, the camshaft **must** be replaced.

Inspect the camshaft lobes for cracks, scratches, and other damage. Refer to Camshaft Reuse Guidelines for Cummins® Engines with Roller Followers or Roller Tappets, Bulletin 3666052, (/qs3/pubsys2/xml/en/bulletin/3666052.html) for further reuse guidelines.



If the valve camshaft and/or the rocker levers need to be replaced, use the Camshaft Compatibility Matrix to determine whether the camshaft and rocker lever roller combination being installed is compatible.

If the matrix states that the combination of the camshaft and rocker lever roller is “No”, the correct combination **must** be installed using the Valve Camshaft Compatibility Matrix.

The wide rocker lever roller is 17 mm [0.67 in] and the narrow rocker lever roller is 15 mm [0.59 in].

The wide camshaft lobe is 20 mm [0.79 in] and the narrow camshaft lobe is 18.5 mm [0.73 in].

Valve Camshaft Compatibility Matrix	Wide Camshaft Lobe	Narrow Camshaft Lobe
Wide Rocker Lever Roller	Yes	No
Narrow Rocker Lever Roller	Yes	Yes

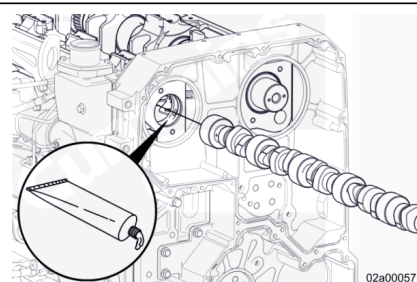
Install

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ CAUTION ⚠

Use caution when installing the camshaft to avoid damaging the camshaft bushings.



Coat the inside of the valve camshaft bushings with Lubriplate™ prior to installing the camshaft.

Use the Valve Camshaft Removal and Installation Kit, Part Number 5299837, to install the camshaft into the cylinder head.

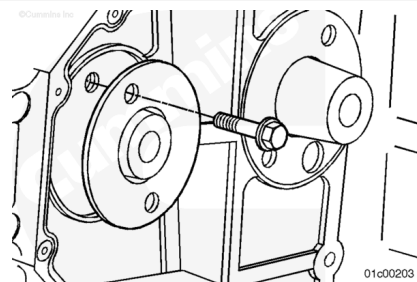
Install the camshaft by hand until three journals are in the cylinder head. Use the Camshaft Installation and Removal Kit, Part Number 5395348, to pull and guide the camshaft into the cylinder head.

Install new o-rings onto the retainer plate.

Install the retainer plate and capscrews.

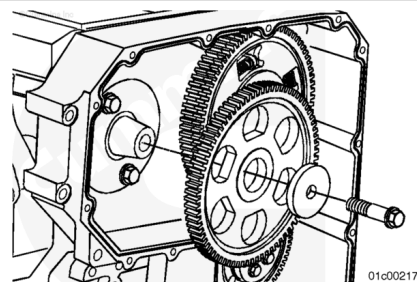
Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]



Finishing Steps

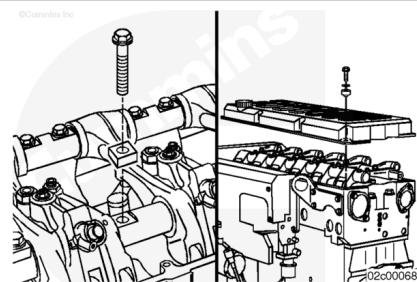
Install the valve camshaft gear. Refer to Procedure 001-088 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html>).



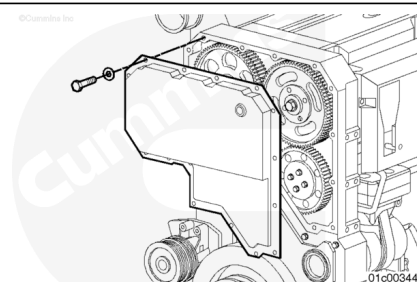
Install the rocker lever shafts. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html>).

Connect the engine brake wiring harness. Refer to Procedure 020-015 in Section 20 (</qs3/pubsys2/xml/en/procedures/10/10-020-015.html>).

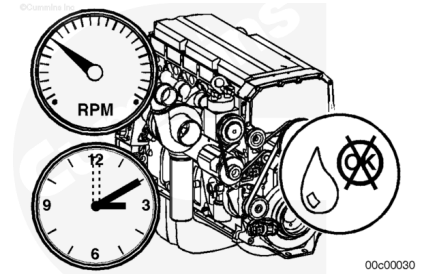
Install the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>) in Section 3.



Install the upper gear cover. Refer to Procedure 001-079 (</qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html>) in Section 1.



Operate the engine until it reaches normal operating temperature. Check for leaks.



Last Modified: 10-Sep-2019
